

# Correlation filter tracking

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## I. INTRODUCTION

In this assignment, the short-term SiamFC tracker was adapted into a long-term tracking framework by using the peak value of the correlation response as an indicator of target presence. Various thresholds were tested to determine the optimal value for detecting target disappearance. The impact of the number of sampled regions on the tracker's re-detection performance was also evaluated. Cases where the tracker successfully re-detected the target were identified and visualized. For the re-detection mechanism, uniform sampling across the entire image was initially employed, followed by an evaluation of Gaussian sampling centered around the last confident target position.

## II. EXPERIMENTS

### A. Short-term vs long-term tracker performance

In the first part of the evaluation, we began by testing the short-term implementation of the SiamFC tracker and then compared it to the modified long-term version. Both trackers were evaluated on all sequences in the provided dataset. The long-term modification builds on the short-term tracker by introducing a thresholding mechanism based on the correlation response. When the peak response falls below a predefined threshold, a re-detection step is triggered. This step samples 50 candidate positions uniformly across the entire image and selects the one with the highest correlation response. The re-detection process remains active until a response exceeding the threshold value of 4 is found.

The comparison of the trackers is presented in Table I. While the precision of the long-term variant decreased slightly—possibly due to re-detection being unnecessarily triggered in some frames where the target was still present, leading to suboptimal localization—the recall and, consequently, the F-score improved significantly. This improvement is expected, as the short-term tracker typically fails to recover once the target leaves the field of view, whereas the long-term extension is able to re-detect and continue tracking.

| Tracker    | Precision | Recall | F-score |
|------------|-----------|--------|---------|
| Short-term | 0.5982    | 0.3251 | 0.4212  |
| Long-term  | 0.5817    | 0.4524 | 0.5090  |

Table I

PERFORMANCE OF SHORT-TERM AND LONG-TERM TRACKING.

### B. Searching for the optimal value of confidence threshold

As previously mentioned, the maximum value of the correlation response was used as a confidence score. In the initial experiments, the threshold was set to 4 based on observations: the correlation peak typically dropped from values around 6–9 to approximately 2–3 when the target was lost. The chosen threshold thus fell midway between these ranges. In this section, we systematically evaluate a range of thresholds around this preliminary value to identify the most effective setting.

On Figure 1 we can observe the results at different thresholds. We can see that the best overall F-score is achieved with the thresholds set to 4 or 5. For further evaluations we opt for 4 since it has the highest F-score and this way we have

less unnecessary re-detections. We can see that the precision actually keeps growing with a higher threshold, however the recall plummets.

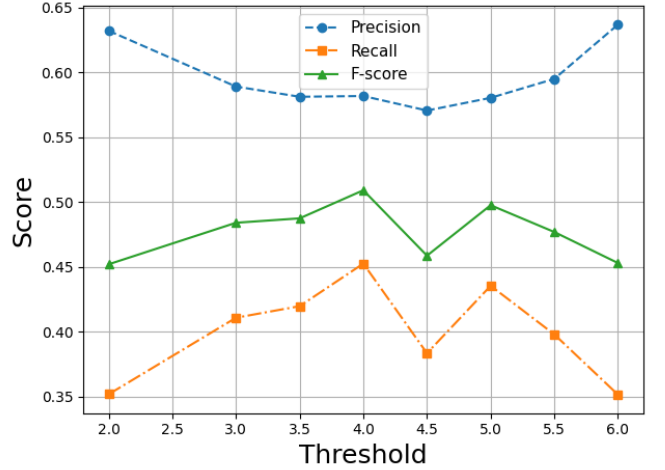


Figure 1. Comparison of performance at different thresholds

### C. Testing different number of randomly sampled regions

Next, we examined how the number of sampled regions during re-detection impacts the tracker's performance. For each sequence, we recorded the total number of frames required for successful re-detection, then averaged these values across all sequences. The results are presented in Figure 2. Interestingly, the number of frames required for re-detection does not consistently decrease as the number of sampled regions increases, which would be expected if the tracker functioned perfectly. This may be due to the fact that sampling more regions increases the likelihood of false positives, which can mislead the tracker and ultimately delay successful re-detection.

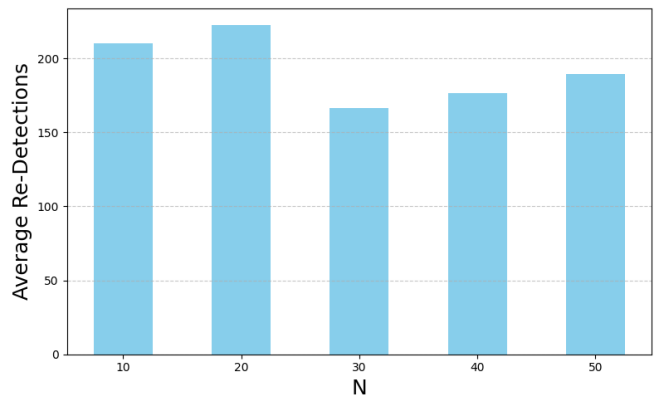


Figure 2. Average number of re-detection frames for different numbers of sample points

#### D. Visualization

Here we present a visualization of the target re-detection process using our tracker, shown in Figure 3. In the first image, the target is occluded by a traffic billboard, leading the tracker to lose the target and instead assign the position to a visually similar patch. In the second image, as the car reappears from behind the obstruction, our re-detection mechanism successfully re-identifies and tracks it.



Figure 3. Example of the long term tracker re-detection

#### E. Different sampling methods

We also tried using the gaussian sampling method, around the last confidently tracked point (before the correlation output was below the threshold). For this method we used a standard deviation of 1/10 of the height of the image. The results (Table II) show a slight improvement in precision but a small drop in recall, with the overall F-score remaining nearly the same. This trade-off can be attributed to the localized nature of Gaussian sampling: the tracker becomes more focused on regions near the last known location, which helps avoid false positives from distant, similar-looking regions—hence the improved precision. However when the target re-appears on a completely different part of the image, the tracker with gaussian sampling is more likely to miss it, leading to reduced recall.

| Sampling Strategy | Precision | Recall | F-score |
|-------------------|-----------|--------|---------|
| Uniform           | 0.5817    | 0.4525 | 0.5090  |
| Gaussian          | 0.5971    | 0.4430 | 0.5087  |

Table II

PERFORMANCE COMPARISON BETWEEN UNIFORM AND GAUSSIAN SAMPLING STRATEGIES.

### III. CONCLUSION

In this assignment we successfully modified a short-term tracker to long-term tracker and significantly improved the performance on long-term sequences. We showed that for our implementation of re-detection with correlation response peaks, the optimal threshold was 4. We found out that the number of sampled points does not necessarily mean a faster re-detection. We showed examples of successful re-detection. Lastly we tried 2 different sampling strategies: uniform across the entire frame and gaussian around the last successfully tracked point with a constant standard deviation. We found out that both methods perform similarly.